Project Executable Files

Date	29 June 2026
Team ID	xxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	2 Marks

ProjectExecutable Files

This section contains all the executable and deployment files required for the OptiCrop Smart Agricultural Production Optimization system. The project is developed using Python, Machine Learning, and Flask. All source code, trained model files, dataset, and frontend/backend components are organized properly to ensure smooth execution and evaluation.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt (dependency file)	Yes
4	Dataset (Credit Card Approval Dataset)	Yes
5	Environment configuration file (.env.example if applicable)	NA
6	Deployed application URL (if hosted)	No

7	Executable application (Flask Web Application)	Yes
8	Dockerfile / Containerization setup	NA
9	Model file(model.pkl)	Yes
10	Demo video / Walkthrough	Yes

Step 2: File / Folder Structure

The OptiCrop project follows a modular structure separating dataset, model, frontend, backend, and utilities for better maintainability and scalability.

```
OptiCrop/
│
├── dataset/
│   └── Crop_recommendation.csv
│
├── models/
│   └── model.pkl
│
├── static/
│   ├── css/
│   └── images/
│
├── templates/
│   ├── home.html
│   ├── about.html
│   └── findyourcrop.html
│
├── app.py
├── model_training.py
├── preprocessing.py
├── requirements.txt
├── README.md
└── prediction.py
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed
Login Credentials (Demo)	Not required
Platform / Hosting Provider	Local Flask Server
Repository link	https://github.com/Harshitha-17-stack/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Demo Video Link	(Add your Demo Video Link)

Step 4: Run Instructions

Follow the steps below to execute the OptiCrop project locally:

1. Download or clone the project repository.
2. Open the project folder in Visual Studio Code or PyCharm.
3. Create and activate a Python virtual environment (optional).
4. Install all required dependencies using:

   pip install -r requirements.txt

5. Ensure dataset and model file are in correct directories.
6. Start the Flask application using:

   python app.py

7. Open the browser and visit:

   http://127.0.0.1:5000

8. Enter soil and environmental parameters.
9. Click Predict to get crop recommendation.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction depends on dataset quality	Improve dataset with more real world data
2	Local deployment only	Can be deployed using Render / AWS / Azure
3	No explanation for predictions and Limited Crop dataset	Future work: add explainable AI (XAI) and Expand with region specific datasets.